Hazard Analysis
Software Engineering

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Table 1: Revision History

Date	**Developer(s)**	**Change**
Date1	Name(s)	Description of changes
Date2	Name(s)	Description of changes
...	...	...

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[You are free to modify this template. —SS]

1 Introduction

[You can include your definition of what a hazard is here. —SS]

2 Scope and Purpose of Hazard Analysis

[You should say what **loss** could be incurred because of the hazards. —SS]

3 System Boundaries and Components

[Dividing the system into components will help you brainstorm the hazards. You shouldn't do a full design of the components, just get a feel for the major ones. For projects that involve hardware, the components will typically include each individual piece of hardware. If your software will have a database, or an important library, these are also potential components. —SS]

4 Critical Assumptions

[These assumptions that are made about the software or system. You should minimize the number of assumptions that remove potential hazards. For instance, you could assume a part will never fail, but it is generally better to include this potential failure mode. —SS]

5 Failure Mode and Effect Analysis

[Include your FMEA table here. This is the most important part of this document. —SS] [The safety requirements in the table do not have to have the prefix SR. The most important thing is to show traceability to your SRS. You might trace to requirements you have already written, or you might need to add new requirements. —SS] [If no safety requirement can be devised, other mitigation strategies can be entered in the table, including strategies involving providing additional documentation, and/or test cases. —SS]

Table 2: Failure Mode and Effect Analysis (FMEA)

Failure Mode	Effects of Failure	Causes of Failure	Risk (H/M/L)		Recommended Action	SR	Ref.
			Sev.	Occ.			
Web Server (WS) — SRS S.2.1							
Server down							H-WS-1
Server externally attacked							H-WS-2
Server overloaded from legitimate traffic							H-WS-3
Server loses connection to internet							H-WS-4
Server compromised							H-WS-5
Rogue actor gets access to change server							H-WS-6
Access Control System (AC) — SRS S.2.2							
Wrong user gets in							
Data leak (I/E)							H-AC-1
Password leak (I/E)							H-AC-2
Convenience leak (Superior gives inferior account access without clearance)							H-AC-3
Role category given extra rights							H-AC-4
Wrong person assigned to role category							H-AC-5
Right user locked out							
Forgot password							H-AC-6
Role category given less rights							H-AC-7
Right person not given role							H-AC-8

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Table 2 continued

Failure Mode	Effects of Failure	Causes of Failure	Risk (H/M/L)		Recommended Action	SR	Ref.
			Sev.	Occ.			
Toolbox Broker (BR) — SRS S.2.3							
Knowledge Repository (KR) — SRS S.2.4							
Stored data lost or corrupted	Club data stored since the last backup is lost	Host or server failure, backup failure	H	L	Store backup of system (KR) off the server, on another platform	NFR-KR-2, SR-KR-1	H-KR-1
Item overwritten or deleted by mistake	A design or document is lost	Member error	M	M	Create confirmation prompts for deletions and modifications (e.g. replacing a file if uploading a copy), Access control protects file modification and deletion from all members	FR-KR-5, SR-KR-2	H-KR-2
Team's files visible to another team during the competition year	A team's unreleased design is exposed, contrary to SRS E.3	Access is checked by role only, not by which team owns the file	M	H	During the competition year, individual team files are marked as active and made available only to the owning team and executives	SR-KR-3	H-KR-3
Content cannot be found	Knowledge is lost in practice; work is duplicated	Missing or inconsistent tags, incorrect metadata	M	H	Require a title, a category from a fixed list, and a year on upload	FR-KR-3, SR-KR-4	H-KR-4
Repository unavailable	No saving or loading of stored data	Outage, storage capacity full	H	L	Tell the member when data was not saved	FR-BR-4, SR-KR-5	H-KR-5
Model Analysis System (MA) — SRS S.2.5							
Corruption of input file	A partly read or corrupted model returns wrong analysis results	Bad or interrupted CAD export	H	L	Show an import summary for the member to confirm before analysis or further processing	FR-MA-2, SR-MA-1	H-MA-1
Output display wrong, model analysis display not clear	Member misses a design flaw, bridge state misassessed	Wrong units or scale, errors in PA	M	M	Show units and a scale reference in the 3D view; check usability of the interface in the Winter CD — Usability Report	NFR-MA-2, SR-MA-2, Winter CD usability test	H-MA-2
Timeout	Member gets no response, either retries or gives up	Large model files, heavy load	M	M	Stop processing after MA_TIMEOUT and show error message	NFR-MA-1, SR-MA-3	H-MA-3

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Table 2 continued

Failure Mode	Effects of Failure	Causes of Failure	Risk (H/M/L)		Recommended Action	SR	Ref.
			Sev.	Occ.			
Performance Analysis System (PA) — SRS S.2.5.1							
Calculation API error	No estimate, or a wrong ultimate load prediction	External analysis API unavailable, changed, or returning invalid values	H	M	Show an error instead of an estimate when the API fails or returns an invalid value; have test models to check new API calls against	NFR-PA-1, SR-PA-1	H-PA-1
Output confidence not understood	Estimate is misread or treated as a guaranteed result	Assumptions and limitations hidden, misunderstanding of service	M	L	Show the assumptions and limitations next to each estimate; check usability of the interface in the Winter CD — Usability Report	FR-PA-4, Winter CD usability test	H-PA-2
Rule Compliance System (RC) — SRS S.2.5.2							
Misinterpretation of rules (risk high for high penalty rules)	A non-compliant bridge passes the check and is penalized or disqualified at competition	Rule entered incorrectly, wrong year's rule set active, rule cannot be checked from the model, unit errors cause misrepresented comparison	H	L	Show the year of the rule set used, and highlight rules that require a manual check	FR-RC-5, NFR-RC-1, SR-RC-1	H-RC-1
Material Testing Manager (MT) — SRS S.2.6							
Calculation API error							H-MT-1
Corruption of input file							H-MT-2
Output display wrong							H-MT-3
Output not understandable							H-MT-4
Timeout							H-MT-5
Knowledge Transfer Manager (KT) — SRS S.2.7							
System-Wide (SYS) — SRS S.2.8							
Upgrade causes our code to no longer compile							H-SYS-1
Unchecked, unworking code pushed in haste							H-SYS-2

6 Safety and Security Requirements

[Newly discovered requirements. These should also be added to the SRS. (A rationale design process how and why to fake it.) —SS]

6.1 Functional Requirements

- **SR-KR-2:** The Knowledge Repository shall always prompt the user to confirm an action that will result in deletion or modification of stored files. [H-KR-2]
- **SR-KR-3:** The Knowledge Repository shall require that uploads of current competition year files are uploaded with a current team, limiting file access to only members of that team and executives. [H-KR-3]
- **SR-KR-4:** The Knowledge Repository shall reject an upload that does not include a title, a category selected from a standardized list, and a year. [H-KR-4]
- **SR-KR-5:** The Knowledge Repository shall return an error message whenever an uploaded file cannot be stored or a data retrieval error occurs. [H-KR-5]
- **SR-MA-1:** The Model Analysis System shall display a summary of inputted models to allow users to confirm the details and ensure accuracy and correctness. [H-MA-1]
- **SR-MA-2:** The Model Analysis System shall display the units of the inputted models, along with a scale reference in the 3D view. [H-MA-2]
- **SR-MA-3:** The Model Analysis System shall stop processing a model that has not finished within MA_TIMEOUT, and shall tell the member why processing stopped. [H-MA-3]
- **SR-PA-1:** The Performance Analysis System shall display an error message whenever the API fails, does not respond, or returns an inappropriate value. [H-PA-1]
- **SR-RC-1:** The Rule Compliance System shall display the competition year of the active rules with each result, and shall indicate when manual verification is required. [H-RC-1]

6.2 Non-Functional Requirements

- **SR-KR-1:** Backups of the Knowledge Repository shall be stored on an external platform. [H-KR-1]

7 Roadmap

[Which safety requirements will be implemented as part of the capstone timeline? Which requirements will be implemented in the future? —SS]

8 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

8.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

8.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

8.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

9 Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?
4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?